22. Incorporating text mining into Critical Discourse Studies: A Corpus-Assisted Discourse Study of press representations of climate change in China

Ming Liu and Jingyi Huang

INTRODUCTION

Entering the 21st century, climate change has emerged as a pressing global issue, leading to increased international collaboration to address environmental threats to humanity. Media play an indispensable role in shaping public understanding of climate change, either by highlighting the urgency of the issue and the need for action or by downplaying its significance (Liu & Huang, 2024; Nerlich et al., 2012). They can affect public perception of climate change as a distant problem or an immediate threat requiring collective action. Media representations of climate change tend to be shaped by a number of factors, such as political bias and ideology, cultural values and norms, journalistic practices and norms and geopolitical relations and national interests (Carvalho, 2007; Liang et al., 2014; Song et al., 2021). This has given rise to several studies on media representations of climate change in different socio-political contexts, such as the US (Liu & Huang, 2024; Sonnett, 2022), India (Boykoff, 2010), the UK (Bevitori, 2010; Collins & Nerlich, 2016) and other developed countries (Dirikx & Gelders, 2010; Mocatta et al., 2023).

With the development of corpus linguistics (CL) and Critical Discourse Studies (CDS), an increasing number of studies combine these fields to provide a Corpus-Assisted Discourse Study (CADS) of media representations of climate change (Alexander, 2010; Song et al., 2021). This chapter aims to contribute to this trend by incorporating text mining into CDS to provide a Corpus-Assisted Discourse Study of the representations of climate change in the official English-language newspaper *China Daily* (CD). This study has two specific objectives: (1) to demonstrate how text mining can contribute to a CADS of press representations of climate change in CD and (2) to reveal the specific ways of representing climate change from both synchronic and diachronic perspectives. The next section offers a brief introduction to the research background before turning to CADS (in the third section) and an elaboration of data collection and analytic methods (in the fourth). The penultimate section presents the findings of this study and the final section brings the whole chapter to an end with a brief summary of the findings.

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ENVIRONMENTAL PROTECTION AND COMMUNICATION IN CHINA

Historically a developing nation that emphasised economic growth following its economic reforms in the 1990s, China has shown a shift in its approach to environmental concerns in more recent years (Chen, 2014; Liu & Li, 2017; Liu & Zhang, 2018). Events like the Beijing Olympic Games in 2008, combined with severe air pollution levels, spurred the Chinese government to adopt a proactive environmental stance. In a notable move in 2009, China declared a target to cut carbon intensity by 40 to 45 per cent by 2020, marking its first international policy goal in this area (Heggelund & Nadin, 2017). The 18th National Congress of the Communist Party of China in 2012 saw the introduction of the concept of eco-civilisation and during the 12th Five-Year Plan (FYP) Period (2011–2015), the country implemented several plans, including the National Program on Climate Change, the Work Plan for Controlling Greenhouse Gas Emissions and the 12th FYP for Energy Conservation and Emission Reduction (Qi & Wu, 2013). In 2020, President Xi Jinping announced at the United Nations General Assembly that China would reach the peak of carbon dioxide emissions by 2030 and would achieve carbon neutrality by 2060 (Qi & Dauvergne, 2022), thus aligning with the mainstream institutional discourses of responsible environmental practices adopted by other countries.

In the meantime, the Chinese government also utilises its official English-language news media to represent its climate policies and promote its environmental image to the external world (Liu, 2023; Liu & Huang, 2024; Liu & Li, 2017). Recent years have also seen numerous studies on media representations of climate change in China, with shared findings that constructions of climate change in Chinese media align with the theme of "domestic accomplishment, international responsibilities". For example, Liang et al. (2014) find that Chinese media tend to frame the causes of climate change as anthropocentric and focus on the need for developed countries to be more proactive in addressing climate change issues. Xie (2015) reveals that Chinese newspapers are more likely to focus on domestic mitigation achievement and commitment. Pan et al. (2019) discovered that Chinese news reports in COP21 emphasise China's contribution to COP21 and its proactive role in climate control, tending to portray China as a rising power in need of world recognition. Trading on this body of work, this study provides a critical examination of climate change representations in CD over 20 years illuminating in particular how the official English-language newspaper has adapted to the changing socio-political contexts to represent climate change to a global audience.

CORPUS-ASSISTED DISCOURSE STUDIES

A CADS approach emphasises the "synergy" between CDS and CL (Baker et al., 2008; Partington et al., 2004). The basic assumption is that a balanced combination of the methods and theories of CDS and CL can generate significant and innovative findings, and contribute to the mutual development of both disciplines (Morley & Bayley, 2009). It does not presume pre-determined analytic methods and procedure but requires the analysts, mutually informed by methods and theories of CDS and CL, to identify the most appropriate analytic methods and procedure concerning the data (Baker et al., 2008). However, previous CADS studies have relied on traditional CL methods, such as keywords, collocates and concordance (Poole,

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2022). Despite their proven value, these techniques are not optimal in investigating the overall topic structure and content of texts because they are primarily designed to examine lexical relationships (Murakami et al., 2017).

To meet the challenge of analysing large volumes of texts, some researchers have started using quantitative machine learning and computational linguistics methods to enhance CDS (Liu et al., 2022; Poole, 2022; Törnberg & Törnberg, 2016). For example, Törnberg and Törnberg (2016) have combined CDS with topic modelling to explore the relationship between Islamophobia and anti-feminism on a large online platform. Lee (2019) combines co-occurrence analysis with topic modelling to analyse a corpus of Korean news reports. Additionally, Chen et al. (2021) conducted a semantic network analysis on newspaper articles spanning from 1997 to 2018 to identify prevalent themes. These studies have proven that integrating text mining with CDS can be effective in identifying patterns of language use and underlying ideologies in large collections of text. However, these studies often focus more on text mining and less on detailed linguistic analysis, an issue that Liu (2024) has suggested should be addressed through a synergic combination of text mining and CDS.

DATA COLLECTION AND ANALYTIC METHODS

This study collected news reports concerning climate change from the newspaper CD. This paper was chosen because it is the most influential official English-language newspaper in China and is significant in communicating the voice of the Chinese government to the rest of the world (Guo & Huang, 2002; Liu & Li, 2017). Given the severity of climate change, CD has played a key role in communicating this global crisis and providing detailed coverage of the issue, including its impacts on China and other countries, as well as China's responses to it (Song et al., 2021). It offers readers a proper understanding of climate change and the stances of the Chinese government (Fu & Wang, 2022).

The corpus includes news reports with "climate change" in their headlines. All news texts were extracted from the electronic database *Factiva*, and the time span for data collection extends from 2001 to 2020. Previous studies have indicated that climate change did not receive significant media attention until after the year 2000, with growing interest in the media since the mid-2000s (Schmidt et al., 2013). To ensure the relevance of the collected data, these news reports were further manually examined to eliminate irrelevant texts. The collected news texts were compiled into the CD corpus, which consists of 438 news reports with 281,958 tokens.

The tool KH Coder was used to analyse the CD corpus. KH Coder is capable of performing tasks such as extracting keywords, analysing word co-occurrences, and conducting various types of visual and statistical analyses (Higuchi, 2016, 2017). One of its key features is the ability to recognise multiple concepts within a single document, allowing for a richer understanding of the text (Higuchi, 2016).

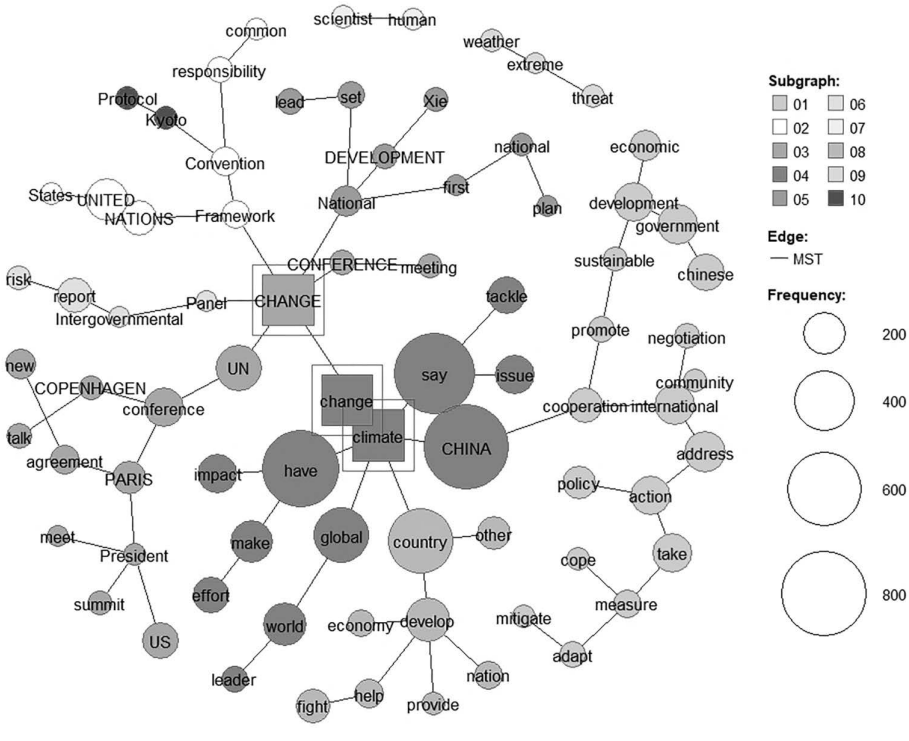
The study started with a co-occurrence network analysis of the term *climate change* within the corpus to offer an overview of the prevalent topics and themes related to climate change in the newspaper. A co-occurrence network examines the frequency with which certain words appear together within a given textual unit, be it a sentence, paragraph, or entire text (Grayson et al., 2016). By analysing the co-occurrence network around specific words, one can infer the topics, content, or frames used to represent a particular issue (Higuchi, 2016). For this analysis, a coding rule was established to focus on the phrase *climate change*, consisting of two

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tokens: *climate* + *change*. The connection strength in a co-occurrence network is measured by the Jaccard coefficient (Liu et al., 2022). KH Coder can also make a distinction between proper and common nouns and present proper nouns in block letters (see Figure 22.1).

A correspondence analysis was further conducted to observe how representations of climate change had shifted over the past 20 years. KH Coder can contribute to the automatic identification of key words for each year (Higuchi, 2016, 2017). The data were analysed year by year, and only content words (such as nouns, verbs, and adjectives) were examined. The diagram represents each year with squares and the significant words for those years with dots. Tokens that are uncharacteristic often plot near the origin (0,0), while those with strong characteristics are positioned further from it (see Figure 22.2). Tokens that occur near the specific year are considered as more frequently used in that particular year (Higuchi, 2016).

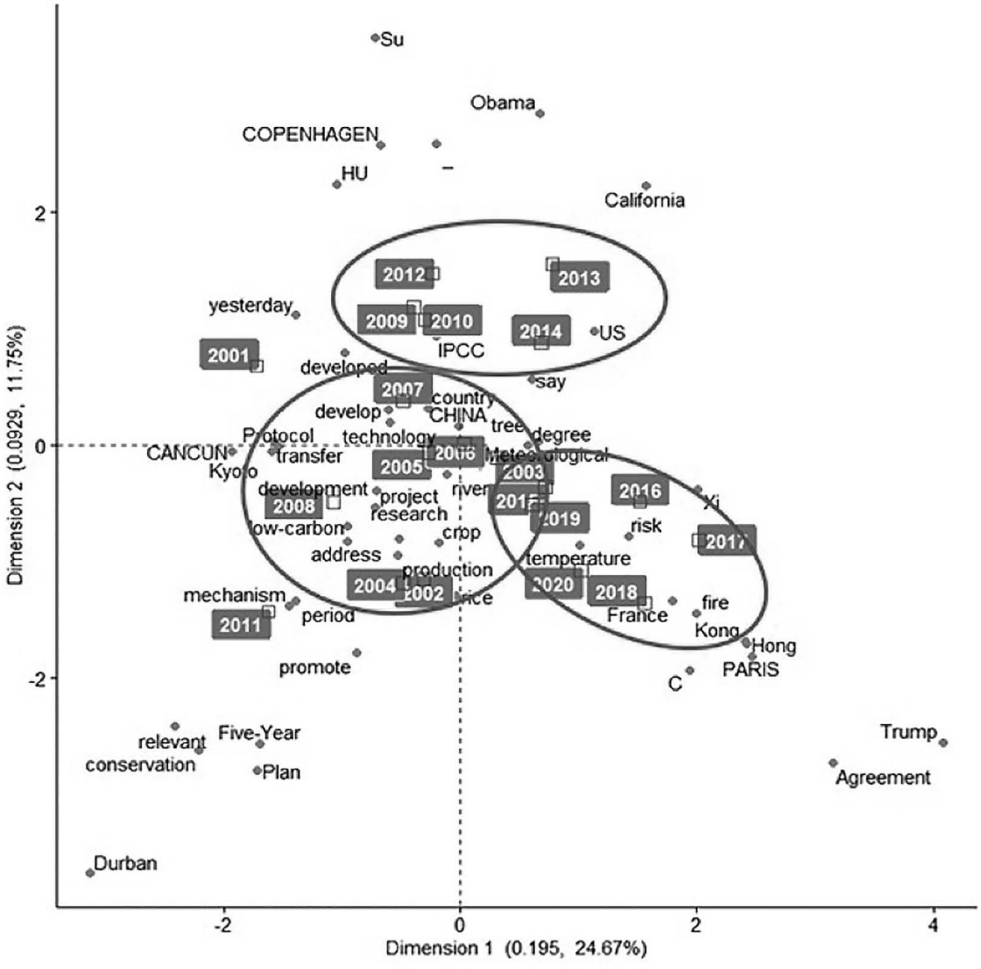
Finally, this study focused on the language CD used to describe the roles of China and the US in the context of climate change. This involved looking at verbs and adjectives that had a strong tendency to co-occur with China and the US, respectively. Coding rules were



Source: Author's own work.

Figure 22.1 *Co-occurrence network of* climate change *in* China Daily

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Source: Author's own work.

Figure 22.2 *Correspondence analysis in* China Daily *each year*

created for both *China (i.e., *China*) and *US (i.e., *US* OR *States*).[1] The strength of their co-occurrence was measured using Log Likelihood (LL) values. These prominent verbs and adjectives were examined in their specific contexts of use to determine how they contributed to constructing the different images of China and the US in the newspaper. They could suggest

[1] "USA" was not taken into account, as it only featured five times in the entire corpus, too few to be of great significance. We chose "States" to represent "United States" as "United" and "States" are taken as two words by KH Coder. Indeed, an examination of the corpus found that all "states" referred to "United States", but "united" is used to refer to not only "United States" but also "United Nations".

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the typical strategies used to construct both China and the US in these news texts, such as predication strategies, that is, the strategies used to attribute a certain property to a social actor or phenomenon (Wodak, 2016).

This study ends with a brief discussion of the findings in terms of the ideology squares of "positive self-presentation and negative other presentation" (van Dijk, 2006), the ideologies of the newspaper and the national interests of China in its representations of climate change. The methodological implications of this study were further drawn to shed light on the analytic potentials of incorporating text mining into CDS.

ANALYSIS OF THE CORPUS

Co-occurrence Network Analysis

Figure 22.1 visualises the co-occurrence network of *climate change* in CD. Since each shade of grey represents a cluster of words that have a strong tendency to co-occur with each other within a sentence, they can suggest different themes. Theme 1 focuses on international cooperation and negotiation in addressing climate change, along with the Chinese government's promotion of sustainable economic development. Theme 2 highlights the United Nations Framework Convention on Climate Change (UNFCCC) and shared responsibilities. Theme 3 underlines the meeting and talk at Copenhagen conferences and new Paris Agreement. Theme 4 underscores China's role as a global leader in tackling the impacts of climate change. Theme 5 relates to China's national development plans. Theme 6 indicates the risk reports released by the Intergovernmental Panel on Climate Change (IPCC). Theme 7 addresses human causes and impacts on climate. Theme 8 discusses the support provided to developing countries in the fight against climate change. Theme 9 highlights the threats posed by extreme weather events. Finally, Theme 10 highlights the Kyoto Protocol.

Therefore, CD tends to frame climate change as a global issue and emphasises the role of the United Nations in addressing it, as seen in Themes 2, 3, 6, and 10, along with the emphasis on international cooperation (Theme 1). On the one hand, the newspaper highlights the functions of the UNFCCC, the new Paris Agreement, and the Kyoto Protocol, as well as the role of the Intergovernmental Panel on Climate Change. On the other hand, it underscores China's efforts to address climate change, as evidenced by Themes 4 and 5. This dual focus is clearly illustrated in Theme 1, which emphasises both international cooperation in managing the impacts of climate change and the Chinese government's promotion of sustainable economic development.

Correspondence Analysis

Figure 22.2 illustrates the results of a correspondence analysis of the distribution of prominent words each year in CD. It reveals that the values for each year are distributed quite closely to one another, suggesting a consistency in CD's framing of climate change over the 20-year span. However, three distinct periods can be discerned based on the distribution of the prominent words, namely Period 1 (2002–2008), Period 2 (2009–2014) and Period 3 (2015–2020). Notably, the years 2001 and 2011 are less closely associated with these periods. The unique positioning of 2001 can be attributed to the limited number of news reports on climate change

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during that year. Conversely, the year 2011 is significant due to the formulation of the National 12th Five-Year Plan, which is acknowledged as the first Five-Year Plan to incorporate a specific climate target and policies on climate change, representing a turning point in China's national policy direction regarding climate change (Qi & Wu, 2013). Accordingly, more media attention was directed at government initiatives and domestic strategies for addressing climate change, as indicated by terms such as mechanism, enhancement, promotion and conservation.

Period 1 focuses on the responsibilities of China and developed countries to assist developing nations, as stipulated by the Kyoto Protocol, which was adopted in Kyoto, Japan, on 11 December 1997, and entered into force on 16 February 2005. Related words include *China*, *developed*, *Kyoto*, *Protocol*, *develop*, *country*, *knowledge*, *transfer*, *research*, *project*, *river*, *crop*, *production*, *address*, *low-carbon*, and *development*. This suggests that CD tends to highlight the impacts of climate change on crop production and the developed countries' responsibility for transferring knowledge to developing countries.

However, Period 2 covers events such as the meeting between the former Chinese President Hu Jintao and former US President Barack Obama, and the activities of the Intergovernmental Panel on Climate Change (IPCC), as indicated by terms like *IPCC*, *US*, *California*, *Hu*, *Copenhagen*, and *Obama*. CD not only underscores the importance of global consensus in climate governance and joint efforts to address the common challenge but also portrays China as an active supporter of international agreements when negotiating global responsibilities for tackling the issue. This may be due to the release of the U.S.–China Joint Announcement on Climate Change in 2014, which was a significant moment in the two countries' joint leadership against climate change. The collaboration with the US also reflects China's commitment to joint efforts in addressing the global issue. Consequently, CD tends to foreground the roles of the US and China in combating climate change. This suggests China's increasingly proactive involvement in international climate politics and negotiations since the Copenhagen Conference in 2009 (Kopra, 2019).

Period 3 is associated with the new Paris Agreement, Donald Trump, and the risks of climate change, as evident from terms like as *Paris*, *Trump*, *Xi*, *Agreement*, *Hong*, *Kong*, *fire* and *temperature*. During this period, CD gives increasing attention to climate risks, as illustrated by the frequent use of the term *risk**. This may be due to the rising occurrences of abnormal climatic phenomena, natural disasters, and conflicts in recent years, as signified by words like *conflict* and *fire*. Such framing emphasises the severity, urgency and necessity of addressing the climate crisis. Specifically, natural disasters in Australia are mentioned, emphasising the global impact of climate change and the need for collective action. These collective actions encompass both global efforts, as seen with the Paris Agreement, and domestic commitments, as indicated by the word *Xi* (referring to President Xi Jinping), which became prominent in 2015 and 2016. The focus on President Xi during this time may be linked to major events, such as his pivotal speech in 2015, which significantly contributed to the conclusion of the Paris Agreement, and developments in 2016. This suggests that the newspaper strongly highlights China's stance on global climate governance and its role in advancing climate action, particularly following COP21 in Paris in 2015.

Therefore, the years 2009 and 2015 represent two "critical discourse moments" (Carvalho, 2005, 2007) that influence how CD depicts climate change. These critical moments coincide with significant political events: the 2009 Copenhagen Summit, which emphasised the principle of "common but differentiated responsibilities and respective capabilities" in addressing

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climate change, and the 2015 United Nations Climate Change Conference in Paris, where the Paris Agreement was signed by nearly 200 nations to mitigate the impacts of climate change.

Analysis of Co-occurring Words of *China and *US

Table 22.1 shows the top 20 co-occurring verbs of *China (including *China*) and *US (including *US* OR *State**) ranked by their Log-likelihood (LL) values.

As Table 22.1 shows, *China tends to co-occur with these verbs which underline the action taken towards climate change, including *address* and *fight*. Examples are as follows:

(1) **China** actively participates in worldwide efforts to **address** climate change, earnestly observes the United Nations Framework Convention on Climate Change.
(CD, 10/30/2008)

(2) The United Nations appreciates **China**'s sustained efforts to **fight** climate change, the UN climate chief said.
(CD, 08/29/2017)

Furthermore, it also underlines the efforts and achievements China has made to address climate change, as can be seen from such verbs as *promote*, *achieve* and *work*. Examples are as follows:

(3) **China** will also make extensive efforts to **promote** afforestation in the urban areas and speed up the building of forest eco-barriers for the cities.
(CD, 11/23/2011)

*Table 22.1 Top 20 co-occurring verbs of *China and *United States in the corpus*

Rank	*China Words	*China Conditional	*China LL	*US Words	*US Conditional	*US LL
1	*say*	705 (0.281)	85.22	*withdraw*	19 (0.028)	87.86
2	*promote*	90 (0.036)	51.86	*sign*	21 (0.031)	35.91
3	*peak*	29 (0.012)	39.17	*pull*	9 (0.013)	35.76
4	*address*	91 (0.036)	34.08	*announce*	17 (0.025)	28.07
5	*achieve*	71 (0.028)	34.00	*ratify*	11 (0.016)	26.40
6	*tell*	60 (0.024)	33.46	*refuse*	7 (0.010)	22.68
7	*have*	671 (0.268)	32.84	*pass*	11 (0.016)	20.81
8	*work*	92 (0.037)	32.47	*overtake*	5 (0.007)	20.36
9	*improve*	63 (0.025)	26.99	*cut*	27 (0.039)	17.88
10	*fight*	60 (0.024)	25.86	*send*	12 (0.017)	17.87

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(4) Guided by the concept of scientific development, **China** is taking a series of concrete measures to **achieve** its notable 20 percent energy intensity reduction target in the year ahead.

(CD, 06/06/2007)

(5) So, **China** is committed to **working** hard for a sustainable development.

(CD, 12/04/2007)

Therefore, these verbs help to construct China as an active agent in addressing climate change. This can also be seen from the strong co-occurring adjectives of *China in CD (see Table 22.2). The top 10 co-occurring adjectives are *top* (59), *economic* (138), *international* (151), *largest* (57), *active* (27), *constructive* (24), *positive* (39), *joint* (44), *11th* (23) and *relevant* (28). Among them, the adjectives *active*, *constructive* and *positive* are typical positive evaluative words used to give a positive evaluation of China's role in addressing climate change, as in the following:

(6) "**China** will continue to play an **active** and **constructive** role at UN conferences to negotiate a post-Kyoto Protocol agreement" to fight climate change, Xie Zhenhua…

(CD, 10/30/2008)

(7) Moreover, **China** always has a **positive** and **constructive** attitude toward international cooperation.

(CD, 11/08/2009)

By contrast, *US tends to co-occur with these verbs which tend to underline unwillingness and refusal, such as *withdraw*, *pull* and *refuse*. A close examination of these verbs in their sentences find that they tend to highlight the US's withdrawal from the international agreement

*Table 22.2 Top 10 strong co-occurring adjectives of *China and *US*

Rank	*China** Words	*China** Conditional	*China** LL	*US** Words	*US** Condition	*US** LL
1	*top*	59 (0.024)	60.46	*former*	21 (0.031)	52.69
2	*economic*	138 (0.055)	54.02	*joint*	24 (0.035)	38.72
3	*international*	151 (0.060)	47.78	*federal*	8 (0.012)	33.09
4	*largest*	57 (0.023)	42.07	*largest*	23 (0.034)	25.96
5	*active*	27 (0.011)	39.65	*biggest*	16 (0.023)	22.90
6	*constructive*	24 (0.010)	38.14	*cooperative*	8 (0.012)	20.12
7	*positive*	39 (0.016)	34.52	*presidential*	5 (0.007)	18.27
8	*joint*	44 (0.018)	30.65	*greenhouse-gas*	4 (0.006)	17.90
9	*11th*	23 (0.009)	30.54	*bilateral*	11 (0.016)	15.47
10	*relevant*	28 (0.011)	26.63	*other*	46 (0.067)	14.79

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and the refusal to fulfil its responsibilities in coping with climate change. Examples are as follows:

(8) On June 1, he announced the **US** would **withdraw** from the Paris accord.
(CD, 08/11/2017)

(9) Washington's decision to **pull** the **United States** out of the Paris Agreement took effect on Wednesday, having given notice on Nov 4, 2019, that it would leave the accord.
(CD, 11/06/2020)

(10) One of the **US** excuses in **refusing** to implement the protocol is that the document shuns 80 percent of the global population.
(CD, 05/24/2007)

Furthermore, even those verbs which suggest the approval and passing of bills are negatively evaluated, such as *sign*, *announce*, *ratify*, *pass*, and *cut*. They also underline the US's failure to fulfil its responsibilities, as in the following:

(11) The **US** delegation doesn't have the mandate to **sign** any legally binding decisions because they don't have a national climate legislation…
(CD, 10/08/2010)

(12) The **US** never **ratified** the 1997 Kyoto Protocol partly because the pact didn't compel China and other developing economies to lower emissions.
(CD, 08/24/2009)

(13) But he didn't expect a complete solution in Copenhagen in December, "because I do not think the **US** Senate will have **passed** the climate change bill by then."
(CD, 09/23/2009)

However, *overtake* is used to emphasise that China has overtaken the US in using clean energy.

(14) This year China will **overtake** the **US** as the world's largest market for wind turbines…
(CD, 12/17/2009)

The negative presentation of the US can also be revealed in the adjectives co-occurring with the US. The top 10 co-occurring adjectives are *former* (21), *joint* (24), *federal* (8), *largest* (23), *biggest* (16), *cooperative* (8), *presidential* (5), *greenhouse-gas* (4), *bilateral* (11) and *other* (46). Among them, *largest* and *biggest* are typical predication strategies used to describe the US as the largest producer of greenhouse gas emissions and the biggest emitter of greenhouse gases.

(15) The **United States**, the **largest** producer of greenhouse gas emissions, refused to accept the Kyoto Protocol to reduce emissions.
(CD, 05/24/2007)

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(16) China is the second **biggest** emitter of greenhouse gases after the **United States**, whose share of global GHGs accounted for 13 percent in 1990 and 18 percent in 2005.
(CD, 06/06/2007)

DISCUSSION AND CONCLUSION

Co-occurrence network analysis reveals that CD emphasises not only global collaboration in addressing climate change but also China's role as a global leader in this effort. It highlights the importance of international agreements and the role of the United Nations in coping with global climate change. Correspondence analysis reveals the diachronic changes in CD's representations of climate change. Period 1 (2002–2008) mainly focuses on the impact of climate change and the responsibilities of developed countries to help developing countries. However, as China adopts a more proactive stance in communicating environmental problems and climate change aligning with the international community discourses (Chen, 2014), Period 2 (2009–2014) and Period 3 (2015–2020) highlight the presidents of the US and China, and the environmental policies of both countries. A detailed analysis of the co-occurring words with *China and *US in their contexts finds that CD tends to construct China as an active participant in addressing climate change issues (see also Wu, 2023), and portrays the US as shirking its climate change responsibilities. This is well illustrated through the analysis of verbs and adjectives co-occurring with *China and *US.

Therefore, this study has revealed the ideological square of "positive self-presentation and negative other-presentation" at work in CD's representations of climate change (van Dijk, 2006). This cannot be adequately explained without referring to the unique role of CD within China's media system and the changing socio-political context. As the most prominent official English-language newspaper in China, CD serves as the mouthpiece of the Chinese government and bears the responsibility of conveying the government's viewpoints to the international community (Guo & Huang, 2002; Mody, 2010). It consistently aligns with the Chinese government's stance and national interests when communicating about climate change (Liu & Huang, 2024). Therefore, it emphasises global collaboration in addressing climate change on the one hand and China's proactive approach to climate change on the other hand. Nonetheless, the representations also vary systematically with the evolving positions of the Chinese and US governments towards climate change. Before 2009, CD emphasised the responsibilities of developed countries to assist developing countries in addressing climate change, thus downplaying its role in this issue. However, after 2009, CD began to highlight the actions of China and the US more distinctly in its representations of climate change, thus constructing a positive image of China and a negative image of the US (Kopra, 2019; Song et al., 2021). Therefore, as the "mouthpiece" of the Chinese government, CD always aligns with the Chinese government and shoulders the responsibility of building a positive image of the Chinese government (Lee et al., 2000; Wu et al., 2015). It helps to communicate the voices of the Chinese government and construct the legitimacy of the Chinese government in global environmental issues (Wu, 2023). Nevertheless, the representations of climate change in CD should not be taken for granted, and a diachronic analysis has revealed that it tended to vary with the changing environmental policies of the Chinese government and Sino–US relations.

In sum, a combination of text mining and CDS can present a more nuanced and focused analysis of the representations of climate change at different levels of discourse (Poole, 2022).

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Unlike traditional corpus linguistic methods such as keyword, collocate, and concordance analyses, the incorporation of text mining can offer alternative ways of processing data based on different units of language (Liu et al., 2022). It can uncover interesting language patterns which may serve as entry points for further detailed linguistic analysis. This aligns well with the aims of CADS (Morley & Bayley, 2009). Additionally, the visual presentation of findings in multiple ways can facilitate analysis and engage readers' interest. Thus, a Corpus-Assisted Discourse Study should transcend traditional corpus linguistic methods to identify the most salient patterns in a more efficient and accurate manner. It is hoped that this study will encourage more research in this direction.

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